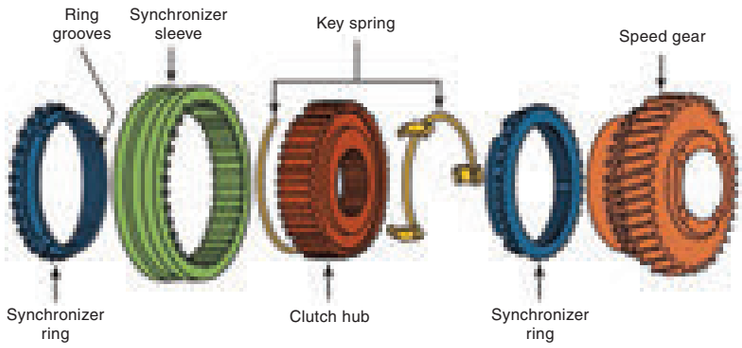



**Figure 3.5.4 Exploded view of a synchronizer unit**

## 3.6. Properties of gear box

1. Gear box should have long life.
2. Noise should be lower in the gear box.
3. Fitting the gears in the gear box should be simple.
4. Replacing the worn out and broken parts should be easy in the gear box.
5. It should transmit the torque and speed according to the requirement.

## 3.7 Gear Ratio

The ratio between the speed of the driver gear to the speed of the driven gear is called as the gear ratio or it is a ratio between number of gear teeth in the driver gear to the number of gear teeth in the driven gear.

$$r_G = \frac{N_2}{N_1} \text{ or } \frac{T_2}{T_1}$$

Gear ratio = clutch shaft speed / main shaft speed

$$\text{Gear ratio} = \frac{speed\ of\ the\ driven\ gear}{speed\ of\ the\ driven\ gear} \text{ or}$$

$$\frac{\text{No of teeth's in driven gear}}{\text{No of teeth's in driven gear}}$$

Where,

$r_G$ = Gear Ratio

$N_1$ = speed of the Driver gear

$N_2$ = speed of the Driven gear

$T_1$ = No of teeth's in driver gear

$T_2$ = No of teeth's in driven gear

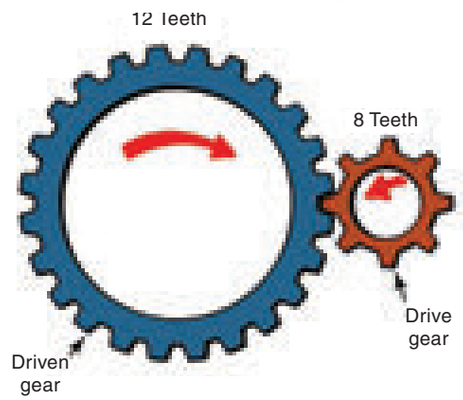



**Figure 3.7 Gear ratio**

## 3.8 Over drive

The method of increasing the speed of propeller shaft more than that of the engine speed is called as the over drive. A view of the overdrive can be seen in Figure 3.8 (a).

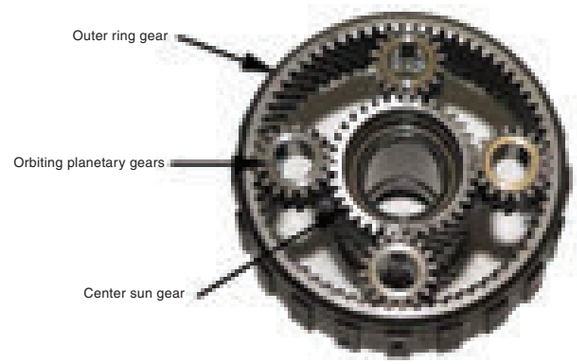



**Figure 3.8 (a) A view of automobile overdrive**



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